

Dietary Recommendation Assistant for Diabetes Prevention

(a) Definition of the Problem

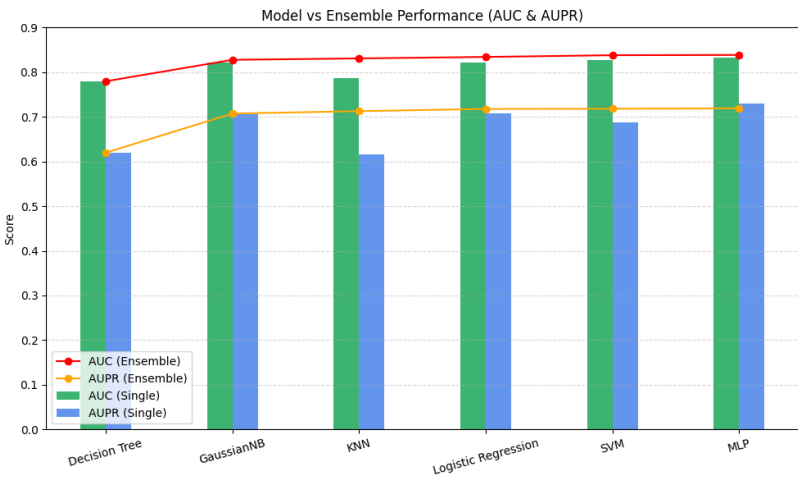
This project addresses the need for personalized dietary planning for people with diabetes. We built a system that predicts diabetes risk using health data (e.g., glucose, BMI, age) and recommends suitable meals based on user preferences.

(b) Rationale of Using a Generative AI Approach or Evaluation

Generative AI is used to extract structured food preferences from user text input using a defined Model Context Protocol (MCP). For example, from a statement like “I love seafood but avoid sugar,” the LLM outputs a structured score for each food category (e.g., +5 for seafood, -4 for sweeteners). This makes user input machine-readable. We also use LLM to generate final recommendation explanations, ensuring the system is interpretable and user-friendly. The MCP ensures output remains consistent and controllable.

(c) Conclusion (with Visual)

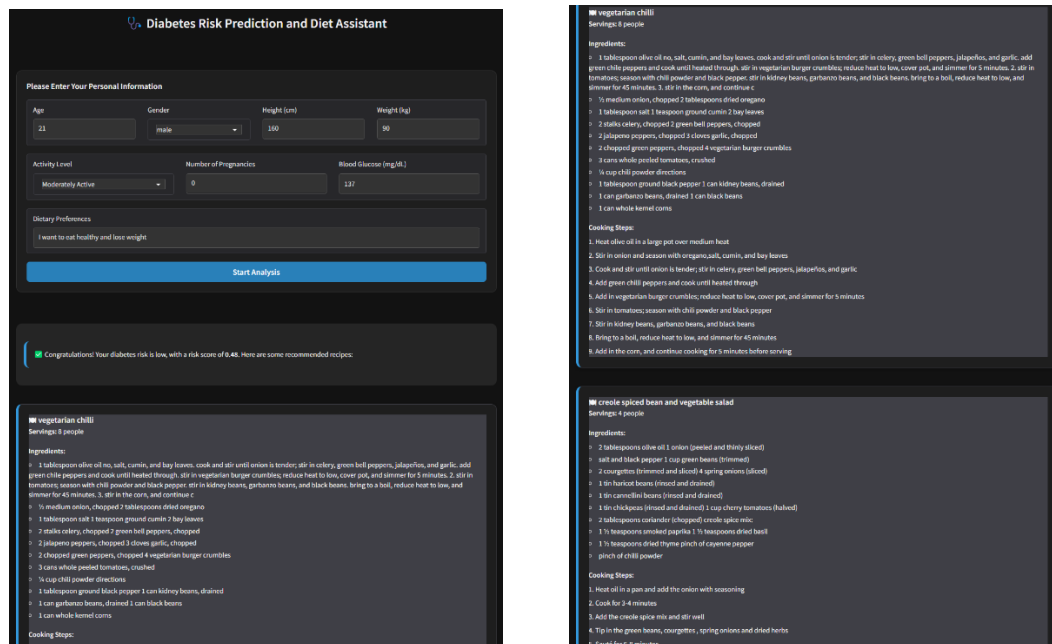
We built a personalized, diabetes-aware recipe recommendation system using machine learning and generative AI. It predicts diabetes risk, understands user preferences, and recommends suitable recipes.



Visual 1: Single Models vs. Ensemble Performance

A supervised machine learning pipeline with model ensembling (e.g., SVM, RF, MLP) ensures robust predictions, which guide recipe recommendations tailored to the

user's condition. Ensemble models outperform single classifiers in AUC and AUPR, providing more reliable predictions.



Visual 2: Application GUI Screenshot

A simple GUI lets users enter preferences, view predictions, and get personalized food suggestions—making the system accessible for real-life use.

(d) Scope for Future Work

- Expand to other conditions like hypertension or weight loss.
- Add filters like allergens, cuisine, and prep time.
- Show calorie/macronutrient estimates dynamically.

(e) Did You Go Beyond Expectation?

Yes, our project went well beyond standard implementation. Here's how:

- Built a robust ML pipeline with soft voting & stacking.
- Integrated LLM + MCP to extract structured preference scores.
- Implemented diabetic-specific filtering with warnings for risky recipes.
- Added a user-friendly GUI to make the system accessible to non-technical users.
- Outputs are clear, structured, and ready for real-world applications.

This project combines traditional ML, structured AI engineering (MCP), and state-of-the-art generative techniques into a cohesive health assistant system.